accurate your measurement, that is, the smaller your error box, the larger the ranges of wavelengths and velocity, and thus the larger the uncertainty in the particle's velocity.

The uncertainty principle governs not only measurements of microscopic objects such as electrons, atoms, and molecules; it also governs measurements of large objects. However, because a large object has large inertia, a measurement's kick will perturb its velocity only slightly. (The velocity perturbation will be inversely proportional to the object's mass.)

The uncertainty principle, when applied to a gravitational-wave detector, says that the more accurately a sensor measures the position of the end or side of a vibrating bar, the more strongly and randomly the measurement must kick the bar.

For an inaccurate sensor, the kick can be tiny and unimportant, but because the sensor was inaccurate, you do not know very well the amplitude of the bar's vibrations and thus cannot monitor weak gravitational waves.

For an extremely accurate sensor, the kick is so enormous that it strongly changes the bar's vibrations. These large, unknowable changes thus mask the effects of any gravitational wave you might try to detect.

Somewhere between these two extremes there is an optimal accuracy for the sensor: an accuracy neither so poor that you learn little nor so great that the unknowable kick is strong. At that optimal accuracy, which is now called *Braginsky's standard quantum limit*, the effect of the kick is just barely as debilitating as the errors made by the sensor. No sensor can monitor the bar's vibrations more accurately than this standard quantum limit. How small is this limit? For a 2-meter-long, 1-ton bar, it is about 100,000 times smaller than the nucleus of an atom.

In the 1960s, nobody seriously contemplated the need for such accurate measurements, because nobody understood very clearly just how weak should be the gravitational waves from black holes and other astronomical bodies. But by the mid-1970s, spurred on by Weber's experimental project, I and other theorists had begun to figure out how strong the strongest waves were likely to be. Roughly 10^{-21} was the answer, and this meant the waves would make a 2-meter bar vibrate with an amplitude of only $10^{-21} \times$ (2 meters), or about a millionth the diameter of the nucleus of an atom. If these